1 Embryonic stem (ES) cells are highly regarded in research because their pluripotency offers the potential for a wide range of therapeutic and research applications. However, ethical concerns have been raised with regards to the source of ES cells.

Scientists have come up with an alternative method of generating pluripotent cells, which is to genetically reprogramme adult cells to an embryonic stem cell-like state. Genetic reprogramming is carried out by using deactivated retroviruses to introduce the genes of four transcription factors into adult cells from a patient (Fig. 1.1).The reprogrammed cells, called induced pluripotent stem (iPS) cells, are specific to the patient from which the adult cells were taken.

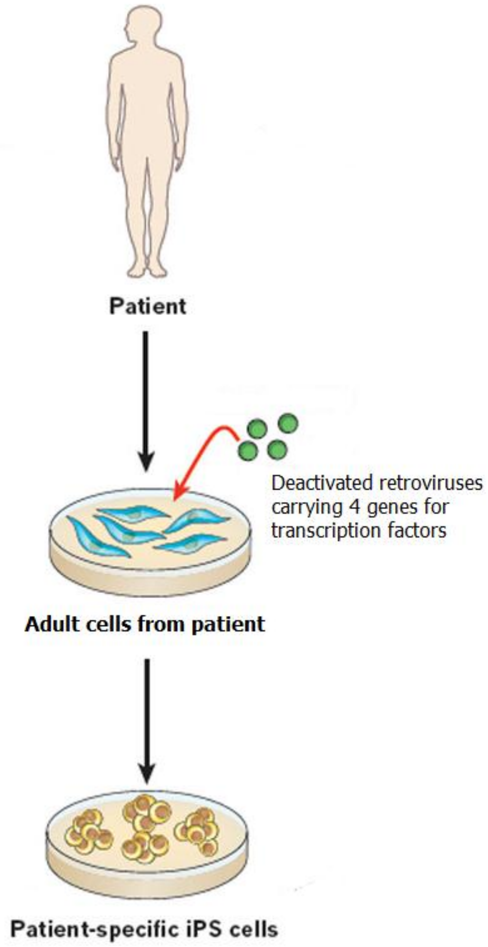

Fig. 1.1

(i) Describe what is meant by the term 'pluripotent'.

It describes the potential of a cell to become/differentiate into any cell type in the adult body but not those of the extra-embryonic membranes; @ 1m [1]

(ii) Suggest **one** reason why the use of iPS cells evades the ethical concerns regarding the source of ES cells.

Ref. to iPS uses adult cells, whereas obtaining ES cells involves the destruction / endangering / risk of embryos;
A! pts
@ 1m

[1]

For Examiner's Use

(iii) Give **three** reasons why deactivated retroviruses are used for genetic reprogramming of the adult cells rather than active retroviruses or liposomes.

***1. deactivated because they will not cause disease/infect other cells /will not become virulent;**

2. retoviruses allow for integration of genes into host genome while liposomes do not;

3. retroviruses are more efficient at delivering genes into the host cell than liposomes because they target specific cells while liposome does not fuse with target cell;

R! ref. to toxicity of liposomes as liposomes not administered *in vivo*.

Both side of ans must be given to award 1 mark except for pt 1.

R! Lysozyme;

***required for full marks**

[3]

(iv) Suggest how the expression of such a small number of transcription factors in adult cells could genetically reprogramme these adult cells to an embryonic stem cell-like state.

1. Each transcription factor can activate the transcription of multiple genes;

2. Each gene in turn could code for a transcription factor which activates/inactivate other genes;

3. transcription factors switched on are those expressed in (pluripotent/ES cells) / transcription factor could also inhibit gene expression;

4. e.g. telomerase gene/ genes which promote cell division/ genes which cause the cell to revert to undifferentiated state;

5. resulting in the synthesis of proteins which are found in ES cell;

6. e.g. genes which result in specialisation (are repressed);

@ 1m, max 3

[3]

Patient-specific iPS cells can be induced to differentiate into the affected diseased tissue as this will allow the patient's disease to be modelled *in vitro*. Using this diseased tissue, potential drugs to treat the disease can be screened, aiding in the discovery of new drugs.

Scientists have managed to obtain iPS cells from cystic fibrosis (CF) patients with a mutation at amino acid position 508 (F508). They have caused these iPS cells to differentiate into diseased human lung tissue, which is now being used to screen for potential drugs to treat cystic fibrosis caused by the F508 mutation (Fig. 1.2)

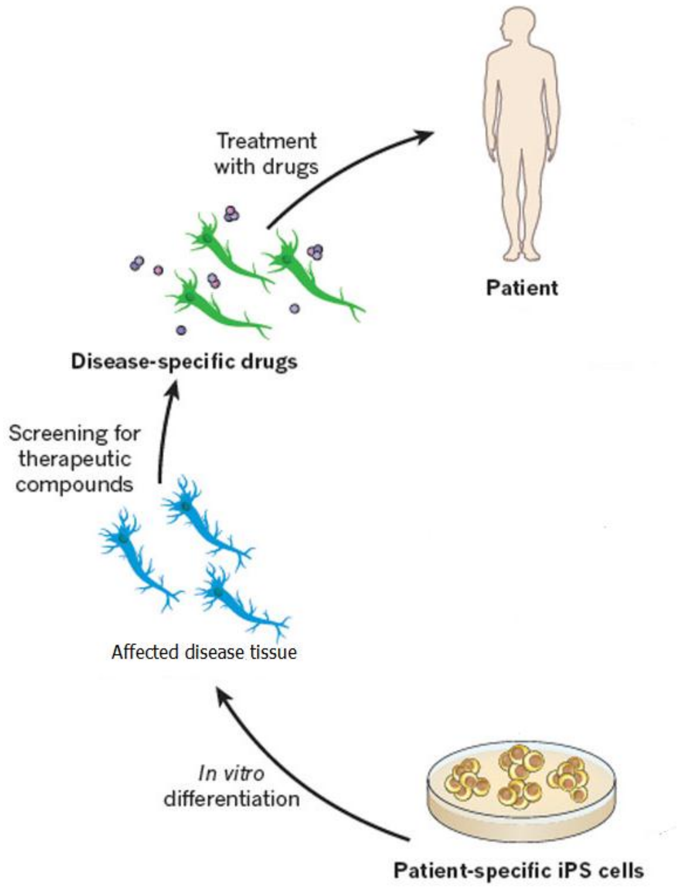

Fig. 1.2

Previously, a cell line comprising of rat cells expressing recombinant human mutant CFTR was used for screening purposes. This cell line was successfully used to identify a drug to treat cystic fibrosis.

(v) Explain the effect of the F508 mutation on the function of the CFTR protein.

1. Mutation results in the deletion of 3 bases / phenylalanine from CFTR protein;

2. resulting in change in tertiary/overall 3D shape change in structure;

3. The CFTR protein is unable to transport chloride ions across the membrane/out of the cell;

[3]

(vi) Suggest and explain **one** advantage of using a drug to treat CF rather than to use liposomes to deliver a functional copy of a CFTR gene.

1. easier to regulate dosage of drug/avoid using high concentrations of liposomes which could be toxic;

2. using the drug avoids the problem that the CFTR gene may not be properly expressed (to form the CFTR protein); [1]

3. Drug is able to travel to many/most parts of body and treat all affected parts whereas liposomes are able to treat only localised areas; (note: liposomes can be injected into the blood stream but the e.g. in the notes relates to the treatment of a localised area)

4. Drug with (specific 3D config) can bind to specific cells while liposome are not specific in targeting cell

Award when either side of point given.
R! Ref. to easier to administer drug than liposomes to patient e.g. drug can be taken orally/by inhaler because liposomes can be administered by inhaler too.
R! Drugs are cheaper than using liposomes to deliver CFTR gene

(vii) Suggest and explain **one** advantage of using diseased lung tissue derived from human iPS cells to screen for CF drugs rather than the use of rat cells expressing recombinant human CFTR.

1. ref. to iPS cells are (human cells and are) more representative of the actual diseased cell than rat cells e.g. rat cells may have different biochemical pathways (which affect CFTR protein or drug)/ OWTTE;

2. ref. to iPS cells are patient-specific and could be used to screen for drugs specific to that patient (i.e. personalised treatment); [1]

[Total: 13]

For Examiner's Use

2 (a) Cysteine proteinase (CP) is an enzyme that is coded by *mir-1* gene, which is found in plants that are naturally resistant to attacks by *Lepidopteran* larvae. *Lepidopteran* larvae feeding on the Black Mexican Sweet (BMS) maize leaf tissue causes crop losses. In an attempt to increase the yield of BMS maize, plant biologists have used genetic technology to transform the wild-type BMS maize callus with *mir-1* gene. The growth of larvae feeding on transformed and non-transformed BMS maize calli was studied for seven days, and the experiments were repeated twice. The results are shown in Fig. 2.1.

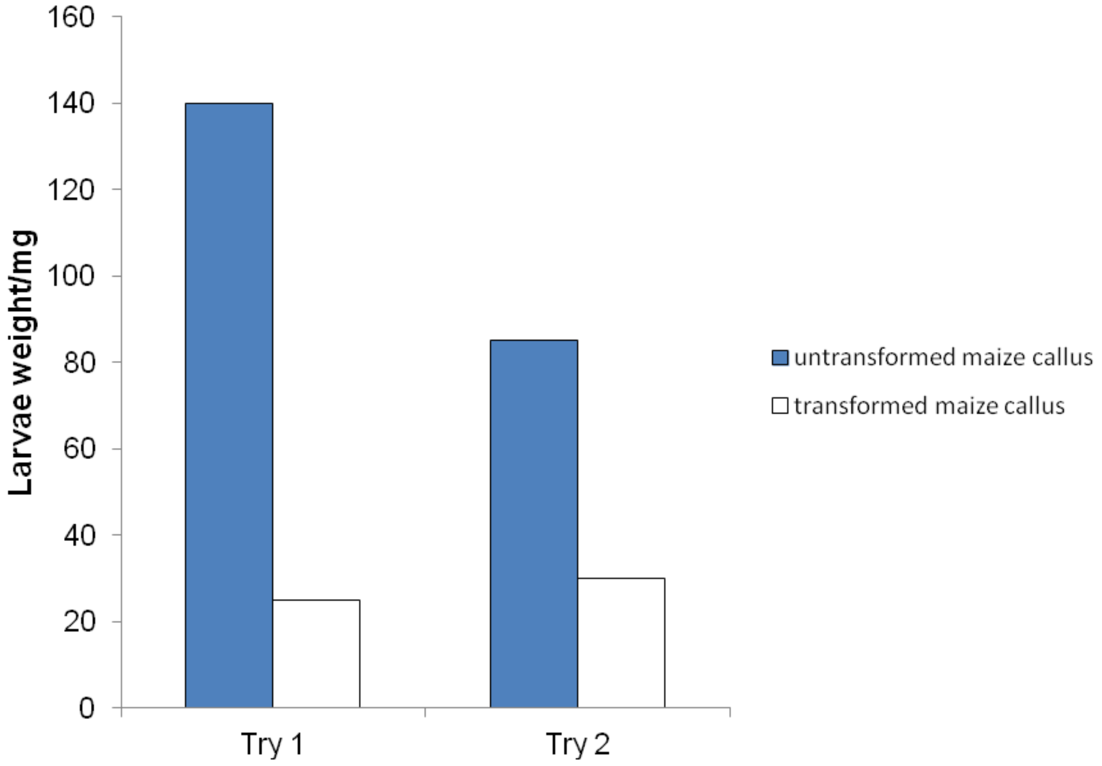

Fig. 2.1

(i) Suggest **one** delivery method that could be used to introduce the *mir-1* gene into BMS maize callus.

1. Ref. to the use of gene gun/soil bacterium/ Agrobacterium/Electroporation; @ 1m

R! liposome (due to presence of plant cell wall) [1]

(ii) With reference to Fig. 2.1, describe and explain the effectiveness of genetically engineered BMS maize with *mir-1* gene on larvae growth.

1. Larvae feeding on transformed BMS maize have a decreased larvae growth / lower weight (as compared to that of the untransformed BMS maize);

2. Ref. to 140 mg as shown in exp. 1 as compared to 25 mg for larvae feeding on transformed maize callus
OR
(85 - 86) mg as shown in exp. 2 for larvae feeding on untransformed maize callus compared to (28-30) mg as shown for larvae feeding on transformed maize callus; @ 1m

3. CP/mir-1 protein was expressed in transformed BMS maize callus <u>and</u> ingested/eaten/consumed by larvae (which slows down/reduces/stunts their growth);

[3]

(b) An *in vitro* experimental study was performed to investigate the effects of transformed BMS plant cells expressing both cysteine proteinase (CP) and *Bacillus thuringiensis* (Bt) toxin on *Lepidopteran* larvae. Table 2.1 shows the relative growth rate and percentage mortality of *Lepidopteran* larvae fed with maize cell lines containing CP, Bt-toxin and a combination of CP and Bt-toxin.

Table 2.1

	CP	**Bt-toxin**	**CP+ Bt-toxin**
Relative Growth Rate/ Arbitrary Units	0.333	0.568	0.220
Percentage Mortality/%	38.0	4.0	50.0

With reference to Table 2.1, describe the effects of maize plant cells expressing CP together with Bt-toxin on the percentage mortality of *Lepidopteran* larvae.

1. **<u>Highest/higher/significant</u> percentage mortality as compared to that of larvae feeding on CP or Bt-toxin maize cells;**

2. **Ref. to figs 50% vs 4 % vs 38 % in table with appropriate units; @ 1 m**

[2]

(c) Both CP and Bt-toxin target the mid-gut lumen. Fig 2.2 shows a section of mid-gut. CP is known to damage the peritrophic protein matrix of mid-gut lumen in the larvae, a structure that protects the mid-gut from microbial infection.

The mechanism of damage by Bt-toxin is different compared to that of CP. Bt-toxin is converted to its active form in the alkaline mid-gut environment, which then binds to the specific mid-gut receptors on the brush border membrane of midgut cells with very high affinity, forming lytic pores that result in cell death.

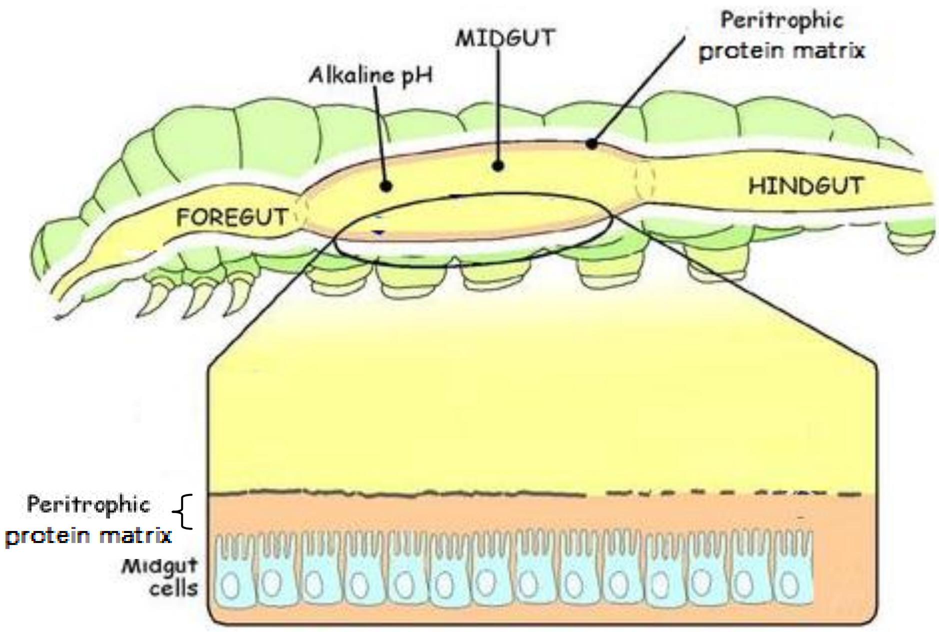

Fig. 2.2

(i) With reference to Table 2.1 and information given above, suggest how CP may enhance the effect of Bt-toxin on *Lepidopteran* larvae.

1. ref. to 4% mortality (relative growth rate 0.568 A.U.) due to Bt toxin alone;

2. CP <u>digests/breakdown</u> proteins found in peritrophic protein matrix, (therefore more porous);

3. Ref. to facilitate the movement of Bt-toxin through the membrane into the brush border membrane of mid-gut lumen /higher accessibility for Bt toxin; [3]

(ii) Over many generations, laboratory populations of *Lepidopteran* larvae have evolved resistance against Bt-toxins. Suggest and explain how the resistance is developed.

***1. Ref. to mutation in genes**

2. The alteration of Bt-toxin binding site on the mid-gut receptors e.g. decreased affinity or reduction in the number of binding sites;

3. Changes in mid-gut environment such that Bt-toxin is not converted to its active form;

4. Rapid regeneration of the damaged mid-gut epithelium; @ 1m
Either pt 1 and 2/3/4

5. (Genotypic/ phenotypic) variation in resistance <u>and</u> selection pressure due to Bt toxin ; [2]

6. Differential reproductive success/fitness;

7. Change in allele frequency (over time);

[Total: 11]

3. Fig. 3.1 below shows a schematic diagram of targeted gene replacement in a mammalian cell involving the gene *Ig M*. Targeted *Ig M* gene replacement is a method of replacing *Ig M* mutated allele (Fig. 3.1a) in the genome in a cell with wildtype allele (Fig. 3.1b) via crossing over, giving rise to the recombinant DNA (Fig. 3.1c). The wildtype allele is carried by a linearised plasmid (Fig. 3.1b) and introduced into the cell.

As shown in Fig. 3.1b, the linearised plasmid contains an antibiotic resistance gene (*Neo gene)*, wildtype alllele with the exons represented by the black boxes, as well as several restrictions sites in between the exons. In this technique, the enhancer of the *Neo* gene on the linearised plasmid was removed prior to the therapy.

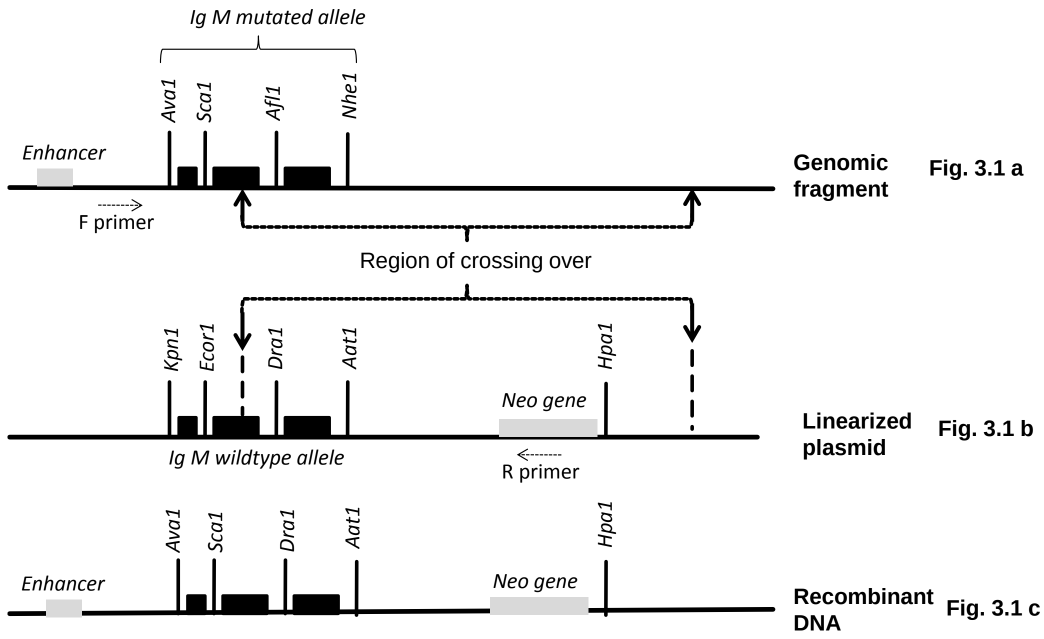

Fig. 3.1

Legend	
Exon	■
Restriction sites on mutated allele	*Ava1, Sca1, Afl1, Nhe1*
Restriction sites on wildtype allele	*Kpn1, Ecor1, Dra1, Aat1*
Naturally occurring restriction site on linearised plasmid	*Hpa1*

i. Explain the significance of the restriction enzyme in bacteria.

1.To (protect) against foreign DNA by digesting/cutting it ; @1m

A! destroy, degrade [1]

ii. Although the mutated and wildtype alleles are homologous regions, different restriction sites are present on them. Explain the presence of these different restriction sites.

1. **(Single) DNA polymorphism / changes in DNA sequences between individual / mutation;**

2. **Creates / destroy / changes restriction sites @1m**

R! RFLP [2]

iii. With reference to Fig 3.1, explain how the *Neo* gene allows the transformed cells with recombinant DNA to be distinguished from transformed cells which did not undergo recombination in their DNA.

1. **Cells that were transformed with subsequent recombination in the chromosome/DNA would survive in the presence of neomycin;**
   A! Neo antibiotic or just a/the antibiotic. R! antibiotic*s*.
2. **Transformed cell without recombination event would not survive**

3. *Neo* gene from vector would come under the control of an active enhancer
4. High expression of *neo* gene confers resistance against neomycin. A! comparison.
5. No /low expression of *neo* gene due to absence of enhancer in linearized plasmid / absence of Neo gene on genomic fragment;

[4]

iv. With reference to Fig 3.1a and Fig.3.1 b, briefly describe another method how *Hpa1* can be used to verify the identity of cells with recombinant DNA.

(Replica plating of cells with recombinant DNA)
***1. (lyse cell) followed by cutting/digesting it with restriction enzyme Hpa1**
R! restriction enzyme added to cells, DNA treated with restriction enzyme
2. separate DNA fragment via gel electrophoresis;
3. stain / Intercalate with ethidium bromide and visualised under UV light;
4. recombinant DNA will show 2 bands while non-recombinant DNA will show band; A! more bands
OR

[4]

***1. (lyse cell) followed by cutting it with restriction enzyme hpa1**
2. separate DNA fragment via gel electrophoresis;
5.Ref to southern blotting / transfer to nitrocellulose membrane;
6.Radioactive probe binds to enhancer
A! any other reasonable specified area or sequence;
7.1 (probe targeting enhancer) short band (recombinant DNA) vs 1 long band (genomic DNA) vs no band (linearised plasmid)
A! other logical patterns depending on the specified probe
Compulsory pt 1

v. The wildtype *IgM* allele was isolated from a genomic DNA library and not a cDNA library before being introduced into the linearized plasmid. With reference to Fig. 3.1 state one evidence which suggest so.

1. allele shown contain introns / non-coding DNA;
R! enhancer

[1]

b) In the gene replacement therapy shown in Fig. 3.1, the site of DNA breakage occurred between restriction sites *Sca1* and *Afl1* as shown in Fig. 3.1a. In another case of therapy, the site of DNA breakage along the mutated allele was unknown. To determine the exact site of crossing over between the genomic DNA and the linearised plasmid, RFLP analysis was carried out on the recombinant DNA. The recombinant DNA was isolated from the transformed cell and amplified using PCR via the forward and reverse primers as shown in Fig. 3.1a and Fig. 3.1b. The amplified DNA products were separated into eight tubes and digested using eight restriction enzymes respectively: *Ava1, Sca1, Afl1, Nhe1, Kpn1, Ecor1, Dra1, Aat1.* The products of digestions were separated using gel electrophoresis and the results are shown in Fig. 3.2.

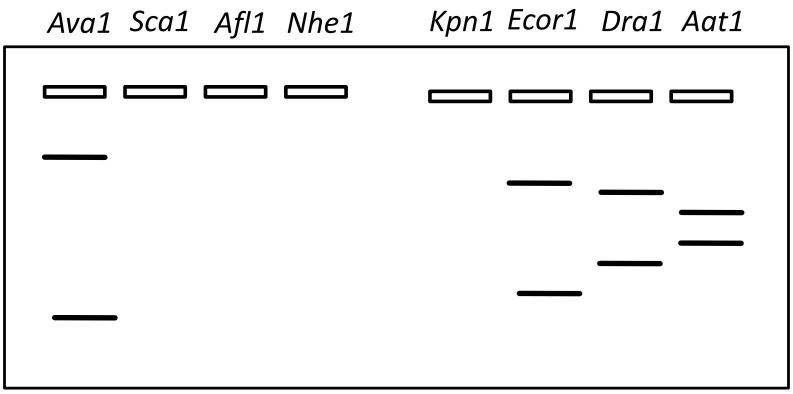

Fig .3.2

(i) With reference to Fig.3.1 and 3.2, describe and explain the band pattern of the restriction digest for the recombinant cell.

1. **presence of 2 restriction fragments each using *Ava1, Ecor1, Dra1* and *Aat1* digest;**
2. **Indicates presence of restriction sites on recombinant DNA.**
3. **digest with *Sca1, Afl1, Nhe1* and *Kpn1* yields no bands,**
4. **indicating absence of restriction sites;**

5. Ref. to explanation linking action of named restriction enzyme/position of a named restriction site (*Ava1, Ecor1, Dra1* and *Aat1)* on the size of the fragments produced;

[3]

(ii) Hence identify the site of DNA breakage where crossing over took place in the mutated allele.

Between *Ava1* and *Sca1;*

[1]

[Total: 16]

For Examiner's Use

Planning Question

4 Electroporation is a method which is routinely used to introduce foreign DNA into *E. coli* cells. This involves subjecting a mixture of bacteria and plasmid to a brief electric pulse. You are required to plan, but not carry out, an investigation into the effect of voltage on the efficiency of transformation by electroporation.

In this investigation, the recombinant plasmid used to transform the *E. coli* cells is made up of two components: the plasmid vector and the gene of interest. pGFP is a plasmid with two marker genes, GFP and *Amp*^{R}, and a number of restriction sites throughout the plasmid as shown in Fig. 4.1 below. GFP is a marker gene coding for the green fluorescent protein. Expression of GFP in bacteria will cause bacteria to fluoresce under UV light. The gene of interest is the *ras* proto-oncogene, which has been inserted into the plasmid using the restriction enzyme *Bmt*I.

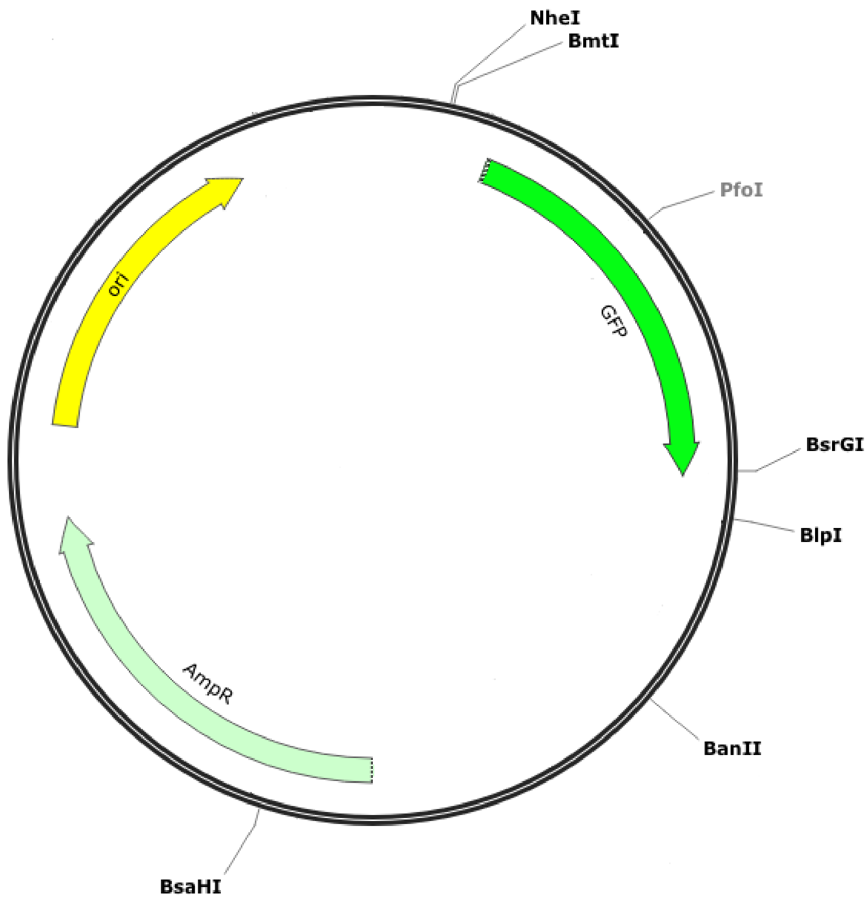

Fig. 4.1

An electroporator is a machine used to carry out electroporation. It is possible to vary the voltage and length of the electrical pulse with this machine. The range of voltage used is typically 1400 – 2000 V and the length of electrical impulse between 4-6 ms. A voltage which is too high will cause irreversible changes in membrane permeability. The mixture of bacteria and recombinant plasmid, which must be kept on ice prior to electroporation, is put into a cuvette (Fig. 4.2) which is then placed into a chamber in the electroporator (Fig. 4.3).

After electroporation, the mixture of bacteria must be quickly resuspended in SOC broth (a type of growth medium) and incubated at 37°C for 1 hour for the surviving cells to recover. This is then followed by plating the bacteria on agar and incubation at 37°C for 16-24 hours. The plates can then be examined to calculate transformation efficiency.

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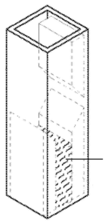

Fig. 4.2

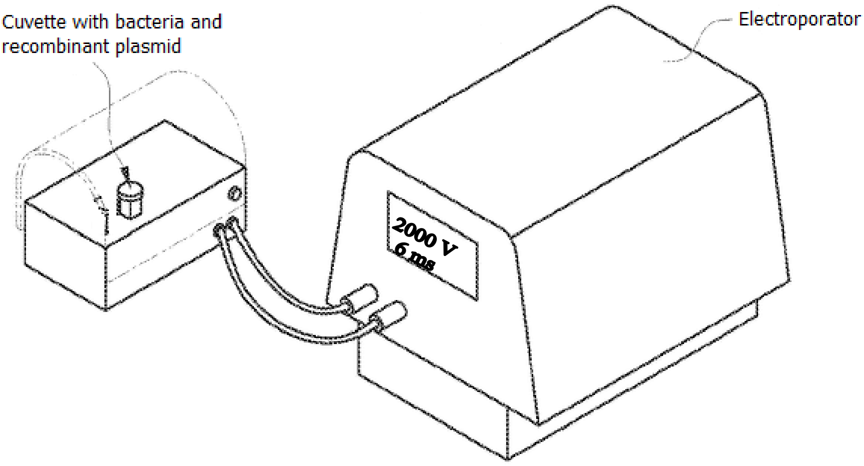

Fig. 4.3

Transformation efficiency can be calculated using the following equation: -

$$\text{Transformation efficiency / cfu } \mu g^{-1} = \frac{\text{Total number of colonies on agar plate}}{\text{Amount of DNA plated (in } \mu g)}$$

[cfu: colony forming unit]

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you must use:

- Microcentrifuge tubes of **100 μl** *E. coli* cells in electroporation buffer
- Microcentrifuge tubes of **1 μl** recombinant DNA (concentration of 0.1 μg / μl)
- Microcentrifuge tubes of **1 μl** sterile water
- Ice bucket with ice
- Electroporator
- Sterile cuvettes for electroporation (max. volume of 150μl)
- Agar plates with ampicillin
- Sterile spreader for spreading bacterial culture on agar plate
- SOC broth
- Incubator
- UV transilluminator (a device which emits UV light)
- UV shields which provide protection from UV light
- A variety of micropipettes, sterile micropipette tips and sterile 2 ml microcentrifuge tubes

Your plan should have a clear and helpful structure to include

- an explanation of the theory to support your practical procedure
- a description of the method used, including the scientific reasoning behind the method, a control experiment and any recommended safety measures
- an explanation of the dependent and independent variables involved
- relevant, clearly labelled diagram/s showing selection of the recombinant bacteria on plates
- how you will record your results and ensure that they are as accurate and reliable as possible
- proposed layout of results tables and graphs with clear headings and labels

For Examiner's Use

- the correct use of technical and scientific terms

[Total:12]

Mark Scheme

1 Theoretical consideration or rationale of the plan to justify the practical procedure

(a) As voltage increases, transformation efficiency increases up to a point where membrane is irreversibly damaged, leading a drop in transformation efficiency as voltage further increases /
At voltages which are too high or too low, the bacteria's ability to take up DNA by transformation becomes reduced, and at high voltages, many/most bacteria will die.

(b) Only transformed bacteria will be able to grow/form colonies on the agar plates with ampicillin. These colonies should also fluoresce under UV light.

2 At least 5 different voltages (between 1400 to 2000 V), regular spacing between intervals

3 (a) Constant volume of *E. coli* cells (eg. 100μl) and
(b) constant volume of recombinant DNA (eg. 1μl)

4 Specify time for incubation of bacteria and recombinant plasmid in ice (1 - 20 minutes). A! incubate together or separately.

5 Specify fixed time for electroporation (4 – 6 ms)

6 Specify fixed vol of SOC (>2ml) and incubate at 37°C for 1 hr

7 (a) Incubate the plates at 37°C
(b) for fixed no. of hours between 16-24hr R! overnight

8 Control experiment – (a) E. coli cells + sterile water subjected to electroporation (no colonies observed)
(b) subjected to same conditions as mixtures with plasmid / repeat exp/ OWTTE

9 Count number of colonies which fluoresce under UV light

10 (a) Replicates
(b) At least two repeats

11 Tabulation:
(a) Ref to colony count
(c) Average transformation efficiency/cfu μg^{-1} under different voltages/V tabulated with correct headings and units

12 Plot graph (with trend line) of ave. transformation efficiency/ cfu μg^{-1} against voltage/ V A! ECF for colonies and units.

13 Drawing of Petri dish with fluorescent colonies labelled

14 Two safety precautions / risk hazards identified

(a) Wear gloves when handling *E. coli* to avoid contact with microorganisms
(b) Wipe table with antiseptic solution (such as ethanol/chlorox) to prevent growth of harmful microorganisms
(c) Use the UV shields to cut down on exposure to UV light
(d) Proper disposal/treatment of contaminated materials or equipment using sterilizer/ autoclave to prevent growth of harmful microorganisms
(e) Handle (plug of) electroporator with dry hands to prevent electrocution.

Max 10 for 2 – 14

15 The correct use of technical and scientific terms

(a) transient pores
(b) ref. to transformed cells being resistant to ampicillin/being able to grow on ampicillin

For Examiner's Use

Free-response question

5 (a) Describe and explain how a bacteria can be used to mass produce human insulin without the use of any DNA libraries. [7]

1. using isolated mRNA for insulin (from human pancreatic cells as template) forming cDNA using reverse transcriptase/reverse transcribe / OWTTE
2. DNA linkers (containing the restriction site) added to cDNA using (DNA ligase)
3. digest plasmids (containing lac*Z* and amp^R or any antibiotic marker genes) and the cDNA with the a/same RE to produce sticky ends. (A! reference to plasmids cut within lacZ by RE)
4. mix the cDNA and the plasmids together and let the sticky ends anneal by H bonds by complementary base pairing
5. Add DNA ligase to covalently join the fragments with phosphodiester bonds (covalent bond to form recombinant plasmids)
6. transform bacteria / electroporation / heat shock
7. plate the transformed bacterial cells onto nutrient agar plate with ampicillin and X-gal added into the medium
8. select colonies which are white in colour as these are the ones with recombinant plasmid (R! bacteria appears white)
9. due to inactivation of Lac z gene
***10. selected bacteria grown (in large quantities) in a suitable medium and bacteria secrete insulin which is extracted, (purified and packaged for use)**

@1m (pt 1 – 9 max 6; pt 10 needed for full marks)

(b) Discuss the advantages and the limitations of the Polymerase Chain Reaction. [5]

Advantages: (@1m, max 4)

1. **Produces large numbers of copies in a relatively short span of time;**
2. **Cell-free method of DNA replication/requires no cleanup of unwanted cellular debris or vector DNA;**
3. **Shown to (efficiently) amplify targets up to 35kb in length;**
4. **A specific process that targets only the desired DNA to be copied, (hence starting material does not have to be purified DNA)**
5. **Only requires a trace amount of DNA;**

Limitations: (@1m)

6. **Need to first know the nucleotide sequence of at least one short DNA sequence on each side of the region of interest in order to synthesise the PCR primers;**
7. **Some errors can be introduced into the amplified DNA copies at low but significant frequencies / Due to lack of a built-in proofreading activity in *Taq* polymerase, PCR produces a higher than normal frequency of replication errors;**

(c) Discuss the implications of the Human Genome Project, including the benefits and difficult ethical concerns for humans. [8]

Benefits max 4

- **Genetic testing (max 1)**

1 The HGP has allowed for discovery of genes/alleles/polymorphisms associated with human diseases/ understanding genetics basis of disease;
2 This allows for improved diagnosis of disease / genetic testing;
3 Earlier detection of genetic predispositions to disease (eg. breast cancer, cystic fibrosis, Alzheimer's disease);
4 It also allows scientists to design drugs to target a specific gene/protein (ie. rational drug design) ;

- **Pharmacogenetics (max 1)**

5 The HGP allows scientists/doctors to know which genes/alleles/polymorphisms affect a person's response to a drug / genetic differences affect the way we react to a drug;
6 It is (theoretically) possible to tailor drugs/treatments to fit patient's genome for greater efficacy / avoiding dangerous side effects / personalized medicine based on personal genotype;

- **Gene therapy**

7 (Since gene sequences are now readily available and we better understand which genes/alleles/polymorphisms are associated with which diseases,) it is possible to use gene therapy to treat certain diseases;

- **Risk assessment of individuals upon exposure to toxic agents**

8 (The HGP has allowed for discovery of genes/alleles/polymorphisms associated with resistance/susceptibility to radiation/carcinogens, and hence) an individual's genome/polymorphisms can be used as a means of risk assessment to evaluate health risks of individuals who may be exposed to radiation or carcinogens;

- **Understanding human evolution (max 1)**

9 (The HGP allows for the discovery of alleles/polymorphisms that can be) used to study evolution through germline mutations in lineages ;
10 Migration/lineage of different population groups can be studied based on female genetic inheritance (using alleles/polymorphisms in mitochondrial DNA) or male genetic inheritance (using alleles/polymorphisms on Y chromosome);
11 (The HGP allows for the discovery of alleles/polymorphisms that can be) used to compare breakpoints in the evolution of mutations with ages of populations and historical events;

- **DNA forensics (max 1)**

(The HGP allows for the discovery of alleles/polymorphisms that can be used to:)
12 Identify potential suspects whose DNA may match evidence left at crime scenes / Exonerate persons wrongly accused of crimes ;
13 Identify crime / catastrophe victims ;
14 Establish paternity and other family relationships ;
15 Identify endangered /protected species (as an aid to wildlife officials) (could be used for prosecuting poachers) ;
16 Match organ donors with recipients in transplant programs ;
17 Detect bacteria/other organisms that may pollute air/water/ soil/ food ;
18 Determine pedigree for seed/livestock breeds ;
19 Authenticate consumables such as caviar/wine/fish ;

Concerns max 4
<u>Ethical and social concerns</u>

i) Privacy and confidentiality of genetic information.
20 **The issue of who owns (and controls) genetic information – individual or company/researcher/government;**

ii) Fairness in the use of genetic information by insurers, employers, courts, schools, adoption agencies, and the military, among others.
21 **The issue of how insurers/employers/courts/schools/adoption agencies/military may request for/use DNA testing/have access to personal genetic information to discriminate people based on their genomes;**

iii) Psychological impact and stigmatization due to an individual's genetic differences.
22 **It is unclear how personal genetic information affects an individual/society's perceptions of that individual / how genomic information affect members of minority communities/ref to Psychological impact and stigmatization due to an individual's genetic differences;**

iv) Reproductive issues (max 1)
23 **There is an issue of whether healthcare personnel are properly counseling parents about the risks and limitations of genetic technology**
24 **The reliability/usefulness of fetal genetic testing has not been verified (in many cases);**
25 **To-be parents may have to make difficult decisions of whether to terminate pregnancy due to presence of genetic disorder (especially one for which there is currently no cure or treatment for) ;**

v) Uncertainties associated with gene tests (max 1)
26 **The issue of whether testing should be performed when no treatment is available/treatment is extremely expensive and the patient cannot afford it;**
27 **The issue of whether parents have the right to have their children tested for adult-onset diseases, (as there is potential for conflict between a parent's choice and a child's welfare);**
28 **There is also the related issue of who has the right to determine whether newborns/others who are incapable of valid consent (eg. mentally incapacitated) should undergo genetic screening;**
29 **The genetic tests may only indicate a probability/chance/not a certainty of a particular polymorphism/allele being associated with a disease or condition/disease developing;**

vi) Conceptual and philosophical implications regarding human responsibility, free will vs genetic determinism, and concepts of health and disease. (Max 1)
30 **The issue of to what extent is a person's behaviour influenced by his genome (instead of the environment/choices)**
31 **The issue of defining what is considered a disorder and what is within the limits of normality.**

vii) Commercialization of products including property rights (patents, copyrights, and trade secrets) and accessibility of data and materials. (Max 1)
32 **The issue of whether genes/DNA sequences can be patented by companies/individuals;**
33 **Patents could impede the development of diagnostics (tests)/therapeutics (treatments) by third parties because of the costs associated with using patented research data;**
34 **Designer drugs developed through pharmocogenomics / genetic tests / treatments discriminates against the poor as they may be expensive/unaffordable; @1 mk, max 4**

pt24/ 29 mark once

[Total: 20]